



HW 2.1: Having studied the ground state energy of the Hydrogen atom with the variational method, we extend it to some more interesting systems. We will estimate the ground state energy of atoms with more than one electron using the variational method. Consider an atom with nuclear charge Z and two electrons. The Hamiltonian of this system is given by

$$H(x) = -\frac{\hbar^2}{2m} (\nabla_1^2 + \nabla_2^2) - \frac{Ze^2}{4\pi\epsilon_0} \left(\frac{1}{r_1} + \frac{1}{r_2} \right) + \frac{e^2}{4\pi\epsilon_0 |\mathbf{r}_1 - \mathbf{r}_2|}, \quad (1)$$

where $\mathbf{r}_1$ and $\mathbf{r}_2$ are the position vectors of the two electrons, and $r_1 = |\mathbf{r}_1|$, $r_2 = |\mathbf{r}_2|$. The first term represents the kinetic energy of the electrons, the second term represents the attraction between the electrons and the nucleus, and the last term represents the repulsion between the two electrons. Using the variational method, we can estimate the ground state energy of this system by choosing a trial wavefunction of the form

$$\psi(\mathbf{r}_1, \mathbf{r}_2) = \frac{Z_{\text{eff}}^3}{\pi a_0^3} e^{-Z_{\text{eff}}(r_1+r_2)/a_0}, \quad (2)$$

where Z_{eff} is an effective nuclear charge that we will vary to minimize the energy expectation value, and a_0 is the Bohr radius ($a_0 = \frac{4\pi\epsilon_0\hbar^2}{me^2}$).

- Calculate the expectation value of the Hamiltonian $\langle \hat{H} \rangle$ using the trial wavefunction $\psi(\mathbf{r}_1, \mathbf{r}_2)$, and find the value of Z_{eff} that minimizes this expectation value.
- Use this value of Z_{eff} to estimate the ground state energy of the atom for $Z = 1$ (H^-), $Z = 2$ (He), and $Z = 3$ (Li^+) in eV.
- Is an anionic hydrogen atom (H^-) possible, i.e., is it a bound state?
- What about for a muonic atom ($m \simeq m_\mu \simeq 200 \times m_e$)? Can a muonic hydrogen anion ($2\mu^-p$) exist as a bound state?



You may find the following integrals useful:

$$\begin{aligned} \int_0^\infty dr r e^{-2Z_{\text{eff}}r/a_0} &= \frac{a_0^2}{4Z_{\text{eff}}^2}, \\ \int_0^\infty dr r^2 e^{-2Z_{\text{eff}}r/a_0} &= \frac{a_0^3}{4Z_{\text{eff}}^3}, \\ \int d^3r_1 \int d^3r_2 \frac{e^{-2Z_{\text{eff}}(r_1+r_2)/a_0}}{|\mathbf{r}_1 - \mathbf{r}_2|} &= \frac{5\pi^2 a_0^5}{8Z_{\text{eff}}^5}. \end{aligned} \quad (3)$$



HW 2.2: Variational Method for Excited States

- a) Show that the variational method can be used to estimate the energy of the first excited state of a quantum system, namely, prove the following upper bound to the energy of the first excited state

$$E_1 \leq \frac{\langle \psi | \hat{H} | \psi \rangle}{\langle \psi | \psi \rangle}, \quad (4)$$

where ψ_0 is the ground state wavefunction and ψ is a trial wavefunction that respects $\langle \psi_0 | \psi \rangle = 0$ (you can assume that we know the ground state $|\psi_0\rangle$ and ground energy E_0 already and that the energy eigenvalues are non-degenerate).

- b) **(Bonus)** Apply this generalized variational method to the quantum harmonic oscillator to estimate the energy of the first excited state. Use a trial wavefunction of the form

$$\psi(x) = xe^{-\alpha x^2}, \quad (5)$$

where α is a variational parameter. Calculate the expectation value of the Hamiltonian

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \frac{1}{2} m \omega^2 x^2, \quad (6)$$

and find the value of α that minimizes this expectation value. Compare your result with the exact energy of the first excited state, $E_1 = \frac{3}{2} \hbar \omega$.

Homework 2 solutions

HW 2.1: The expectation value of the Hamiltonian is given by

$$E_{\text{trial}} = \frac{\langle \psi | \hat{H} | \psi \rangle}{\langle \psi | \psi \rangle}. \quad (7)$$

To calculate this expectation value, we need to evaluate the expectation value of the entire Hamiltonian as well as the normalization of the trial wavefunction. I have already given you the normalization constant of the trial wavefunction, but we can check that it is indeed normalized using some of the integrals provided:

$$\langle \psi | \psi \rangle = \int d^3 r_1 \int d^3 r_2 |\psi(\mathbf{r}_1, \mathbf{r}_2)|^2 = \int d^3 r_1 \int d^3 r_2 \left(\frac{Z_{\text{eff}}^3}{\pi a_0^3} e^{-Z_{\text{eff}}(r_1+r_2)/a_0} \right)^2 \quad (8)$$

$$= \left(\frac{Z_{\text{eff}}^3}{\pi a_0^3} \right)^2 \left(\int d^3 r_1 e^{-2Z_{\text{eff}}r_1/a_0} \right) \left(\int d^3 r_2 e^{-2Z_{\text{eff}}r_2/a_0} \right) \quad (9)$$

$$= \left(\frac{Z_{\text{eff}}^3}{\pi a_0^3} \right)^2 \left(4\pi \int_0^\infty dr_1 r_1^2 e^{-2Z_{\text{eff}}r_1/a_0} \right) \left(4\pi \int_0^\infty dr_2 r_2^2 e^{-2Z_{\text{eff}}r_2/a_0} \right) \quad (10)$$

$$= \left(\frac{Z_{\text{eff}}^3}{\pi a_0^3} \right)^2 \left(4\pi \frac{2a_0^3}{8Z_{\text{eff}}^3} \right) \left(4\pi \frac{2a_0^3}{8Z_{\text{eff}}^3} \right) = 1. \quad (11)$$

So all is well and we can proceed with the denominator equals 1. It doesn't matter anyway since any constant we factor out of the trial wavefunction would have canceled in the expectation value anyway, but at least this way I can restrict our discussion to the numerator of ?? only.

To keep things organized, let us split the calculation into three pieces: 1) kinetic energy, 2) the potential energy due to the nucleus, and 3) the electron-electron repulsion energy:

$$E_{\text{trial}} = \langle T \rangle + \langle V_{\text{nucleus}} \rangle + \langle V_{\text{ee}} \rangle. \quad (12)$$

This problem is very similar to E&M problems with two charges in a central potential, except we will encounter some exponentials from our trial wavefunction. Let us also define for convenience the following constant:

$$C = \frac{Z_{\text{eff}}^3}{\pi a_0^3}. \quad (13)$$

Let us start with the kinetic energy:

$$\langle T \rangle = -\frac{\hbar^2}{2m} C^2 \int d^3 r_1 \int d^3 r_2 e^{-Z_{\text{eff}}(r_1+r_2)/a_0} (\nabla_1^2 + \nabla_2^2) e^{-Z_{\text{eff}}(r_1+r_2)/a_0} \quad (14)$$

$$= -2 \times \frac{\hbar^2}{2m} C^2 \int d^3 r_1 \int d^3 r_2 e^{-Z_{\text{eff}}(r_1+r_2)/a_0} \left(\nabla_1^2 e^{-Z_{\text{eff}}(r_1+r_2)/a_0} \right) \quad (15)$$

$$= -2 \times \frac{\hbar^2}{2m} C^2 \left(\int d^3 r_1 e^{-Z_{\text{eff}}r_1/a_0} \left(\nabla_1^2 e^{-Z_{\text{eff}}r_1/a_0} \right) \right) \left(\int d^3 r_2 e^{-2Z_{\text{eff}}r_2/a_0} \right) \quad (16)$$

where the first equality uses the fact that this integral was symmetric over $r_1 \leftrightarrow r_2$. The last factor is easy to evaluate since it is literally the same integral we did for normalization, so the last parenthesis is just $1/C$. The first integral requires the Laplacian:

$$\nabla^2 e^{-Z_{\text{eff}}r/a_0} = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d}{dr} e^{-Z_{\text{eff}}r/a_0} \right) = \frac{Z_{\text{eff}}^2}{a_0^2} \left(\frac{2a_0}{Z_{\text{eff}}r} - 1 \right) e^{-Z_{\text{eff}}r/a_0}. \quad (17)$$

The first parenthesis is then

$$\frac{Z_{\text{eff}}^2}{a_0^2} 4\pi \left(\frac{2a_0}{Z_{\text{eff}}} \int dr r e^{-2Z_{\text{eff}}r/a_0} - \int dr r^2 e^{-2Z_{\text{eff}}r/a_0} \right), \quad (18)$$

which can be evaluated using the integrals provided gives

$$\left(\int d^3 r_1 e^{-Z_{\text{eff}} r_1 / a_0} \left(\nabla_1^2 e^{-Z_{\text{eff}} r_1 / a_0} \right) \right) = -\frac{Z_{\text{eff}}^2}{a_0^2}. \quad (19)$$

Therefore, the kinetic energy expectation value is

$$\langle T \rangle = 2 \frac{\hbar^2}{2m} C^2 \left(\frac{\pi a_0}{Z_{\text{eff}}} \right) \left(\frac{1}{C} \right) = \frac{\hbar^2}{m} \frac{Z_{\text{eff}}^2}{a_0^2}. \quad (20)$$

This makes sense, the kinetic energy is positive and inversely proportional to the Bohr radius squared. It increases as we decrease the size of the atom.

Now onto the potential energy due to the nucleus. This is a straightforward integral application of the integrals provided. Again, because the sum of the potential for the two electrons is symmetric in r_1 and r_2 , we can just calculate one integral and multiply by 2:

$$\langle V_{\text{nucleus}} \rangle = -2 \times C^2 Z \alpha \left(\int d^3 r_1 \frac{e^{-2Z_{\text{eff}} r_1 / a_0}}{r_1} \right) \left(\int d^3 r_2 e^{-2Z_{\text{eff}} r_2 / a_0} \right) \quad (21)$$

$$= -2C^2 Z \alpha \left(4\pi \frac{a_0^2}{4Z_{\text{eff}}^2} \right) \left(\frac{1}{C} \right) = -2\alpha \frac{Z Z_{\text{eff}}}{a_0}. \quad (22)$$

NOTE: this Z prefactor is different from the variational parameter Z_{eff} (it is the nuclear charge, e.g., $Z = 2$ for Helium, which specifies the nuclear potential). This result also makes sense, the potential energy from the nucleus-electron attraction is negative and acts to keep the system bound.

Now the last and most fun term: the electron-electron repulsion energy. Out of the bat, we know that the integrals showing up here will be nastier since we cannot separate the r_1 and r_2 integrals due to the $|\mathbf{r}_1 - \mathbf{r}_2|$ in the denominator. Life is too short to do this integral by hand, so the hint provided will get us halfway there, we just need to put the integrals in the right form:

$$\langle V_{\text{ee}} \rangle = C^2 \int d^3 r_1 \int d^3 r_2 e^{-Z_{\text{eff}}(r_1 + r_2)/a_0} \frac{\alpha}{\sqrt{r_1^2 + r_2^2 - 2r_1 r_2 c_\theta}} e^{-Z_{\text{eff}}(r_1 + r_2)/a_0} \quad (23)$$

where we have expanded $|\mathbf{r}_1 - \mathbf{r}_2|$ in terms of an angle θ between the two vectors, with $c_\theta = \cos \theta$.

Note that we will have two integrals over solid angles Ω_1 and Ω_2 , but because of rotational symmetry we can choose the z -axis along $\mathbf{r}_1$ so that the integral over Ω_1 is trivial and just gives 4π . The integral over Ω_2 can then be written as an integral over c_θ from -1 to 1 and an integral over the azimuthal angle ϕ from 0 to 2π , which gives another factor of 2π . We have a lot of freedom here to choose our variables because we are integrating over all of $R^3 \times R^3$. Putting all this together, we have

$$\langle V_{\text{ee}} \rangle = 8\pi^2 \alpha C^2 \left(\int_0^\infty dr_1 r_1^2 \int_0^\infty dr_2 r_2^2 e^{-2Z_{\text{eff}}(r_1 + r_2)/a_0} \int_{-1}^1 dc_\theta \frac{1}{\sqrt{r_1^2 + r_2^2 - 2r_1 r_2 c_\theta}} \right). \quad (24)$$

Using the hint for the triple integral in the parenthesis (straight outta **mathematica**):

$$\langle V_{\text{ee}} \rangle = 8\pi^2 \alpha C^2 \left(\frac{5}{2} \left(\frac{a_0}{2Z_{\text{eff}}} \right)^5 \right) = \frac{5}{8} \alpha \frac{Z_{\text{eff}}}{a_0}. \quad (25)$$

Despite the disturbing fractions the result also makes sense: it ought be positive because it is a repulsive potential. And it increases with Z_{eff} because as we pull the electrons closer to the nucleus, they also get closer to each other, increasing their repulsion. Though, it is intriguing that this effect goes only linearly with Z_{eff} , unlike the other two terms which go quadratically (technically $Z_{\text{eff}} \times Z$ for nuclear term).

Putting everything together, we have

$$E_{\text{trial}}(Z_{\text{eff}}) = \frac{\hbar^2}{m} \frac{Z_{\text{eff}}^2}{a_0^2} - 2\alpha Z \frac{Z_{\text{eff}}}{a_0} + \frac{5}{8} \alpha \frac{Z_{\text{eff}}}{a_0} = \frac{\alpha}{a_0} \left(Z_{\text{eff}}^2 - 2ZZ_{\text{eff}} + \frac{5}{8} Z_{\text{eff}} \right). \quad (26)$$

We used $a_0 = \hbar^2/\alpha m$. Let me clean this up a bit. In particular, I do not like that we will need to plug in the Bohr radius explicitly, so let me take a ratio with a known constant: the Hydrogen atom ground state energy $E_0 = -\alpha/2a_0 = -13.6$ eV (note the minus sign here). Since this is just a constant, we can minimize the ratio instead

$$\frac{E_{\text{trial}}(Z_{\text{eff}})}{E_0} = 2 \left(2ZZ_{\text{eff}} - \frac{5}{8} Z_{\text{eff}} - Z_{\text{eff}}^2 \right). \quad (27)$$

Now to find the optimal Z_{eff} , we take the derivative and set it to zero:

$$\frac{d}{dZ_{\text{eff}}} \left(\frac{E_{\text{trial}}(Z_{\text{eff}})}{E_0} \right) = 2 \left(2Z - \frac{5}{8} - 2Z_{\text{eff}} \right) = 0 \quad \Rightarrow \quad \boxed{Z_{\text{eff}} = Z - \frac{5}{16}}. \quad (28)$$

Fascinating. Naively, one would think that the effective nuclear charge felt by each electron is just the nuclear charge minus 1, because of the screening from the other electron. But our variational calculation tells us that the screening is much less effective, $5/16 = 0.3125$.

b) Using our values of Z_{eff} in ??, we can estimate the ground state energy (rather, an upper limit on it) for $Z = 1, 2, 3$:

- H^- : $E_{\text{trial}} = -12.8$ eV.
- He : $E_{\text{trial}} \approx -77.5$ eV.
- Li^+ : $E_{\text{trial}} \approx -196$ eV.

For comparison, the experimental values are:

- H^- : $E_0 = -14.35$ eV.
- He : $E_0 = -79.0$ eV.
- Li^+ : $E_0 = -198$ eV.

Not bad for such a simple trial wavefunction!

c) Yes, the anionic hydrogen atom is possible as a bound state since our variational calculation gives a negative energy.



While the bound state is indeed possible, our calculation is telling us that it is less bound than hydrogen itself, which has $E_0 = -13.6$ eV (this is not true experimentally, but our variational calculation is just an upper bound after all). The binding energy is defined as the energy required to remove an electron from the atom, so if we try to remove one electron from H^- , we will end up with a hydrogen atom and a free electron, which has a total energy of -13.6 eV. Since -12.8 eV is greater than -13.6 eV, it means that the H^- ion is less stable than a hydrogen atom plus a free electron. Therefore, if our number was correct, the H^- ion could exist as a bound state, but it would quickly decay into a hydrogen atom and a free electron. This is not the case experimentally, but using the real answer of -14.35 eV we see that the even in that case, the second electron is so loosely bound that the anion can be easily ionized (neutralized?) by tiny disturbances.

d) For a muonic atom, the Bohr radius is smaller by a factor of $a_\mu/a_0 = m_e/m_\mu \approx 1/200$, so the energy levels are larger by the inverse factor:

$$E_0^\mu = -\frac{\alpha}{2a_\mu} = 200 \times E_0 \approx -2.72 \text{ keV}. \quad (29)$$

Using the same variational calculation as before, we just need to multiply our previous answers by 200 and we are done. Therefore, anionic muonic hydrogen ($2\mu^-p$) has an estimated ground state energy of -2.56 keV , which is indeed a bound state.



In reality, such an atom is extremely unstable because of 1) the lifetime of the muon, and 2) the fact that the muon can be captured by the proton via the weak interaction, leading to a neutron and a neutrino ($\mu^- + p \rightarrow n + \nu_\mu$). In that case, the remaining system would become a muonic neutronium anion (μ^-n). It is not clear if such an exotic atom could exist as a bound state, but if it does, it is held together *not* by the Coulomb force, but by the interaction between the magnetic moment of the muon and the neutron. Note that this capture process is far less likely for the electron $e^- + p \rightarrow n + \nu_e$ because the electron is orbiting at a much larger radius. The technical term here is that the overlap between the electron wavefunction and the proton is much smaller than the overlap between the muon wavefunction and the proton, so the capture process is much more likely for the muon than for the electron.

HW 2.2:

Okay, just as our proof of the variational method for the ground state, let's expand the expectation value of the Hamiltonian in terms of the energy eigenstates $|\psi_n\rangle$:

$$\frac{\langle\psi|\hat{H}|\psi\rangle}{\langle\psi|\psi\rangle} = \frac{\sum_{n=0}^{\infty} E_n |\langle\psi_n|\psi\rangle|^2}{\sum_{n=0}^{\infty} |\langle\psi_n|\psi\rangle|^2}. \quad (30)$$

The trial wavefunction is orthogonal to the ground state, so $\langle\psi_0|\psi\rangle = 0$, which means that the $n = 0$ term in the sum vanishes:

$$E_1 < \frac{\sum_{n=1}^{\infty} E_n |\langle\psi_n|\psi\rangle|^2}{\sum_{n=1}^{\infty} |\langle\psi_n|\psi\rangle|^2}. \quad (31)$$

Again, adding and subtracting E_1 in the numerator gives

$$\frac{\sum_{n=1}^{\infty} (E_n - E_1) |\langle\psi_n|\psi\rangle|^2}{\sum_{n=1}^{\infty} |\langle\psi_n|\psi\rangle|^2} + E_1, \quad (32)$$

which is greater than E_1 since $E_n > E_1$ for all $n > 1$. And so we are done.

Obviously this statement hinges on ψ , our trial wavefunction, being orthogonal to the ground state. That means we better be sure we know ψ_0 before we can choose ψ . That is rarely the case, but if we have reasons to believe the variational method worked very well for E_0 , then we can iterate. Though, errors will accumulate in this case and the strict inequality may not even hold anymore.